

Which of the following statements is true regarding JOIN operation?

☐ Join operations take two or more relations and return as a result another relation

☐

A join operation is a Cartesian product which requires that tuples in the two relations match (under some condition).

☐ The join operations are typically used as subquery expressions in the FROM clause.

☐ All of the above.

Which among the following is not a type of JOIN operation?

2 points

☐ INNER JOIN

☐ CROSS JOIN

☐ RIGHT OUTER JOIN

☐ DISJOINT

Which among the following is the correct form of CREATE VIEW statement in SQL?

2 points

☐ CREATE VIEW view_name

☐ CREATE VIEW view_name AS <query expression>

☐ CREATE AS VIEW view_name <query expression>

☐ CREATE VIEW AS view_name <query expression>

Which among the following statements are true, when one view may be used in the expression defining another view?

2 points

☐

A view relation v1 is said to depend directly on a view relation v2 if v2 is used in the expression defining v1.

☐

A view relation v1 is said to depend on view relation v2 if either v1 depends directly to v2 or there is a path of dependencies from v1 to v2.

☐ A view relation v is said to be recursive if it depends on itself.

☐ All of the above.

Consider the tables 1 and 2 and answer the questions from 5 to 8

Table 1: Relation **team**

team_id	name	ranking	country
Cric-1	BCCI	1	India
Cric-2	ACB	3	Australia
Cric-3	PCB	10	Pakistan
Cric-4	NCB	9	Nepal
Cric-5	ECB	4	England

Table 2: Relation **players**

team_id	player_id	name	ranking
Cric-1	BCCI-11	Virat	1
Cric-1	BCCI-23	Rohit	3
Cric-3	PCB-22	Azam	2
Cric-2	ACB-12	David	6
Cric-2	ACB-29	Smith	3
Cric-5	ECB-88	Cris	4
Cric-5	ECB-82	Ben	1

Consider the SQL query given below:

2 points

```
SELECT *  
FROM players LEFT OUTER JOIN team  
ON players.team_id = team.team_id;
```

How many rows will be returned by the query?

☐ 5

☐ 7

☐ 9

☐ 8

Choose the correct SQL query which will return the table shown below:

2 points

team_id	name	ranking	country	team_id	player_id	name	ranking
Cric-1	BCCI	1	India	Cric-1	BCCI-11	Virat	1
Cric-1	BCCI	1	India	Cric-1	BCCI-23	Rohit	3
Cric-3	PCB	10	Pakistan	Cric-3	PCB-22	Azam	2
Cric-2	ACB	3	Australia	Cric-2	ACB-12	David	6
Cric-2	ACB	3	Australia	Cric-2	ACB-29	Smith	3
Cric-5	ECB	4	England	Cric-5	ECB-88	Cris	4
Cric-5	ECB	4	England	Cric-5	ECB-82	Ben	1

☐ SELECT * FROM team LEFT OUTER JOIN players

☐ ON players.team_id = team.team_id;

☐ SELECT * FROM team RIGHT OUTER JOIN players

☐ ON players.team_id = team.team_id;

☐ SELECT * FROM team LEFT OUTER JOIN players

☐ WHERE players.team_id = team.team_id;

☐ SELECT * FROM team RIGHT OUTER JOIN players

☐ WHERE players.team_id = team.team_id;

Using the SQL query shown below, a view named **best3** is created.

2 points

```
CREATE VIEW best3 AS
SELECT player_id, name
FROM players
WHERE ranking < 3;
```

Which among the following SQL query will display the table shown below?

player_id	name
BCCI-11	Virat
PCB-22	Azam
ECB-82	Ben

☐ VIEW * FROM best3;

☐ SELECT player_id, name FROM best3 where ranking < 3;

☐ SELECT * FROM best3 WHERE ranking < 3;

☐ SELECT * FROM best3;

2 points

Choose the correct SQL queries that will return the table shown below:

player_id	country
BCCI-11	India
BCCI-23	India
PCB-22	Pakistan
ACB-12	Australia
ACB-29	Australia
ECB-88	England
ECB-82	England

SELECT player_id, country FROM team INNER JOIN players
☐ ON players.team_id = team.team_id;

PROJECT player_id, country FROM players INNER JOIN team
☐ ON players.team_id = team.team_id;

SELECT player_id, country FROM players INNER JOIN team
☐ WHERE players.team_id = team.team_id;

SELECT player_id, country FROM players INNER JOIN team
☐ ON players.team_id = team.team_id;

2 points

Which of the following is/are true for **OUTER JOIN**?

☐ The LEFT OUTER JOIN will return all the tuples from the right table, and the common tuples from the left table.

☐ The LEFT OUTER JOIN will return all records from the left table, and the common tuples from the right table.

☐ The RIGHT OUTER JOIN will return all records from the left table, and the common tuples from the right table.

☐ The RIGHT OUTER JOIN will return all the tuples from the right table, and the common tuples from the left table.

2 points

Which of the following will perform the CROSS JOIN operation between **student** and **course** tables?

SELECT *
☐ FROM student CROSS JOIN course;

SELECT *
☐ CROSS JOIN student, course;

SELECT *
☐ FROM student, course;

SELECT * FROM
☐ CROSS JOIN student CROSS JOIN course;

Which of the following statements are true regarding **views** in SQL?

2 points

☐ A view provides a mechanism to hide certain data from certain users.

☐ Any relation that is not of the conceptual model but is made visible to a user as a “virtual relation” is called a view.

☐ A view is defined using the **create as view** statement in SQL.

☐ All of the above.

Which of the following statements are true in the context of **Materialized Views**?

2 points

☐ Create a physical table containing all the tuples in the result of the query defining the view.

☐ If relations used in the query are updated, the materialized view result becomes out of date.

☐ No need to maintain the view, even if the underlying relations are updated.

☐ Creates a new permanent table in the database.

Which of the following statements are true about views?

2 points

☐ A view is a table with a subset of rows and/or columns from another table.

☐ A view is a stored SQL statement that returns a specific subset of rows and/or columns from another table.

☐ A view does not occupy any physical space except that occupied by the definition of the view.

☐ A view has to be updated every time its parent table is updated.